Patterns Assignment

27 March 2026 16:34

Q:

Sol:- row=1 * ✓

row=2 * * * ✓

row=3 * * * * * ✓

Q:

$\boxed{Col = 2 \times row - 1}$

row = 1

Col = 2×1 - 1 = 1

row=2 ; Col = 2×2-1 = 3

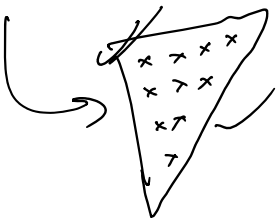
Col = 2×3-1 = 5

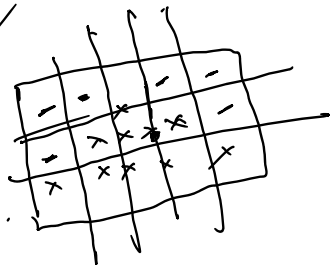
```
for(int row = 1 ; row<=n ; i++)
    for (int j=1; j<n-i+1; j++) Sout(" ");
    for (int j= 1 ; j <= 2×i -1 ; j++){
        Sout("*");
    }
    ctrl +v;
}
```

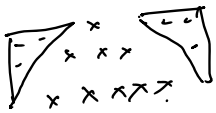
$\boxed{Col = 2 \times row - 1}$

2n-1

2n×1







row=1 * * * * —

row=2 * * * —

* * —

* —

n=4

n

n-1

n-2

n-3

j=1

n-i =3

```
for(int i=1 ; i<=n ; i++){
    for(int j=1; j<= n-i+1 ; j++){
        Sout(*);
    }
}
```

}

row = 1

n - row

i = 3

2 × i - 1

3 - 1

2 × 2 × 1

= 5

Q. row = 1 : 1

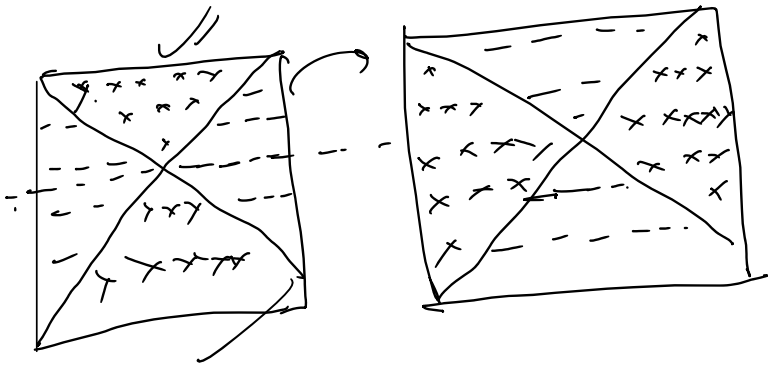
row = 2 : 2 8 2

row = 3 : 3 4 5 4 3

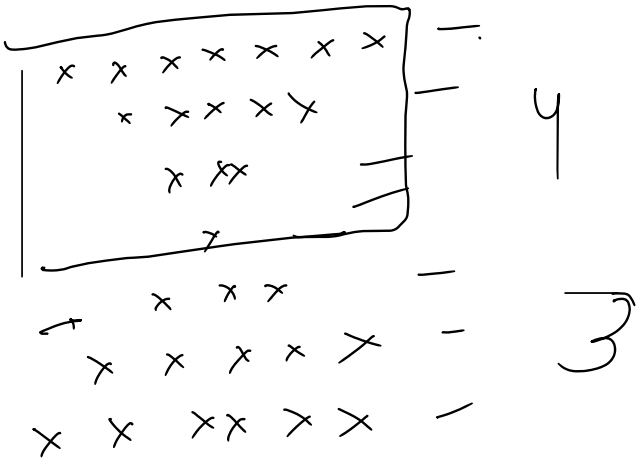
2 × i - 1

```
for (i = 1 → n)
    int count = i;
    for( --- ) ---        //space
    for( int j = 1 ; j <= 2 × i - 1 ; j++){
        Sout (count)
        if( j > i) count --;
        else count ++;
    }
```

Q.



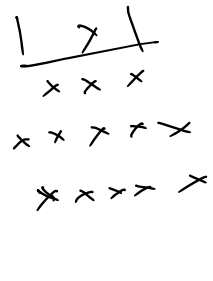
$n = 3$



$4 + 3$
$= 7$

```
for(int i= 1; i<=n+1 ; i++){
    for( int j=1; j<= i ; i++){
        Sout(" ");
        for(int j=1 ; j<= 2* i-1; j++){
            Sout(*);
```

ctrl+c

ctrl+v;

```
}
for(int i=2; i<=n+1; i++){
    ---
    ---
}
```